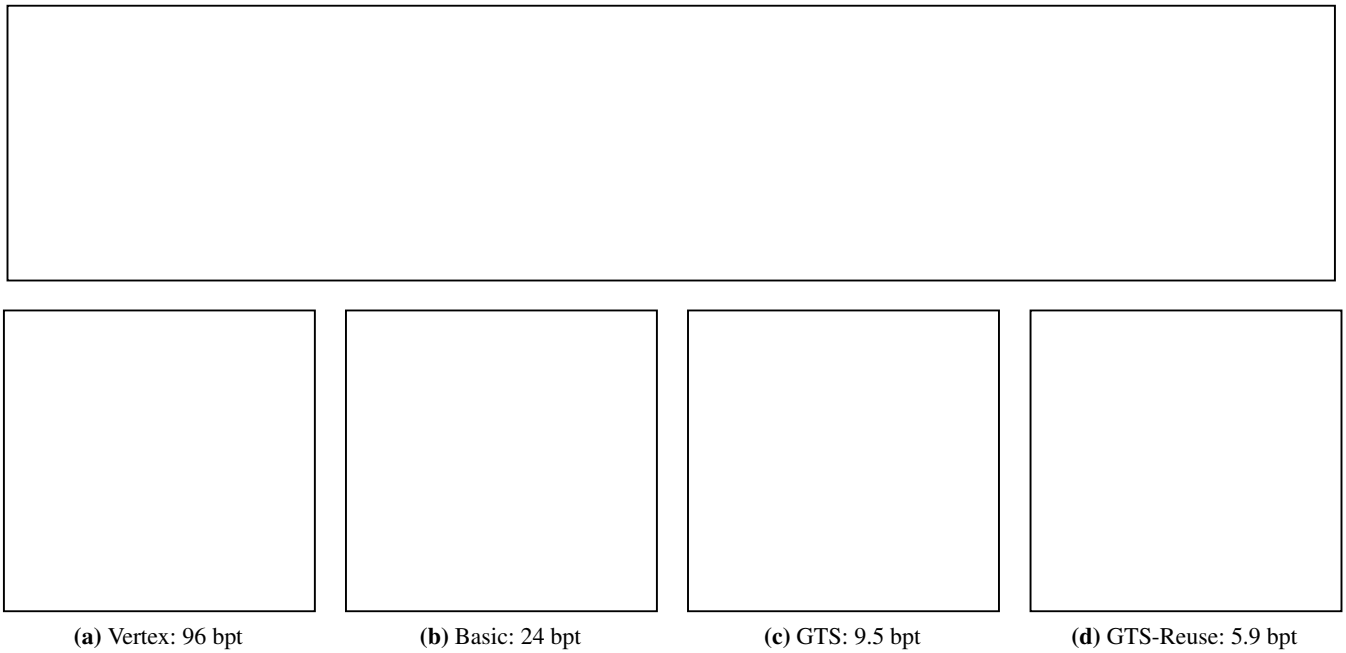
EUROGRAPHICS Workshop on ... '0x
N.N. and N.N.
(Editors)

COMPUTER GRAPHICS *forum*
Volume 0 (1981), Number 0

A Long Title

Bastian Kuth[1] Max Oberberger[2] Quirin Meyer[1]

[1]Coburg University of Applied Sciences and Arts, Germany [2]AMD, Germany



(a) Vertex: 96 bpt **(b)** Basic: 24 bpt **(c)** GTS: 9.5 bpt **(d)** GTS-Reuse: 5.9 bpt

Figure 1: **(a)** Number 1, **(b)** Number 2. **(c)** Number 3. **(d)** Number 4.

Abstract

Here is an abstract. (see https://www.acm.org/publications/class-2012)

CCS Concepts

• ***Computing methodologies*** → *Collision detection;*

1. Introduction

Here we test some text.

Waltz, bad nymph, for quick jigs vex.

Waltz, bad nymph, for quick jigs vex.

Waltz, bad nymph, for quick jigs vex.

`Waltz, bad nymph, for quick jigs vex.`

Waltz, bad nymph, for quick jigs vex.

Waltz, bad nymph, for quick jigs vex.

Waltz, bad nymph, for quick jigs vex.

WALTZ, BAD NYMPH, FOR QUICK JIGS VEX.

2. Previous Work

Test some citations [AG03, PKJ05, MLDH15].

Here is another citation [MSS24].

We will show in Section 3, we can fancy stuff.

Write one line per phrase. Do it like this.

Add a new line to get a new paragraph.

© 0x Eurographics - The European Association
for Computer Graphics and John Wiley & Sons Ltd.

3. Main Part

3.1. Testing some math

A regular formula

$$x^2 + y^2 = z^2 \tag{1}$$

and another one

$$\|\vec{x}\| = \sqrt{\sum_{i=0}^{n} x_i^2 + y_i^2}. \tag{2}$$

Here is something with matrices:

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}^\top. \tag{3}$$

An some inline stuff $\sum a_i = \mathbf{B}$.

4. Results and Discussion

We evaluate the performance of our methods on the meshes of Fig. 2.

Tab. 2 and Tab. 1 compare our optimal method achieved with our other method of Sec. 3 with Gurobi and SCIP against the sub-optimal strip of our other implementation. See Figs. 3, Figs. 4, and Figs. 5 for different types of figures. Here we test the acronyms.

First time we use a graphics processing unit (GPU). Next time, we use a GPU. Now, we even have multiple GPUs.

5. Conclusion and Future Work

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non,

Table 1: Keep a list of symbols in your paper.

Variables used in this paper	
Symbol	**Meaning**
$\mathbf{M}$	model-view matrix
$\mathbf{P}$	projection matrix
$\hat{n}$	a normal vector
V	number of vertices
T	number of triangles
$\vec{l}$	light direction vector
$\mathbf{x}$	point to be lit

Types used in this paper	
Notation	**Meaning**
a	scalar
N	scalar constant
α	angle
$\vec{x}$	column vector
x_i	vector component
$\hat{x}$	normalized vector
$\vec{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}$	column vector
$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,m-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,m-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,m-1} \end{bmatrix}$	matrix
$\vec{a}_{:,j}$	j-th column of matrix $\mathbf{A}$
$a_{i,j}$	element at row i, column j
$\underline{q}$	quaternion
$\mathbf{p}$	point
s_i	i-th element of a series of scalars
$\vec{x}^{[i]}$	i-th element of a series of vectors
$\mathbf{A}^{[i]}$	i-th element of a series of matrices
$\mathcal{A}$	set

Operators used in this paper	
Operation	**Meaning**
$\vec{a}^\top$	transpose, row vector
$\vec{a}^\top \vec{b}$	inner product
$\vec{a} \times \vec{b}$	cross product

Table 2: We compare the optimal Gurobi and SCIP solutions against the sub-optimal stuff. The CPU computation uses one thread per meshlet and was measured on an AMD Ryzen 9 7950X (16C/32T).

	Gurobi	**SCIP**	ETA
computation time	122.46 s	8,122.56 s	10.52 s
GTS render time	1.58 ms	1.58 ms	0.59 ms
GTS restarts	123,784	123,784	185,480
degenerate triangles	2,132	495,136	741,920
additional meshlets	1 / 12412	123	4 / 75,413

© 0x Eurographics - The European Association
for Computer Graphics and John Wiley & Sons Ltd.

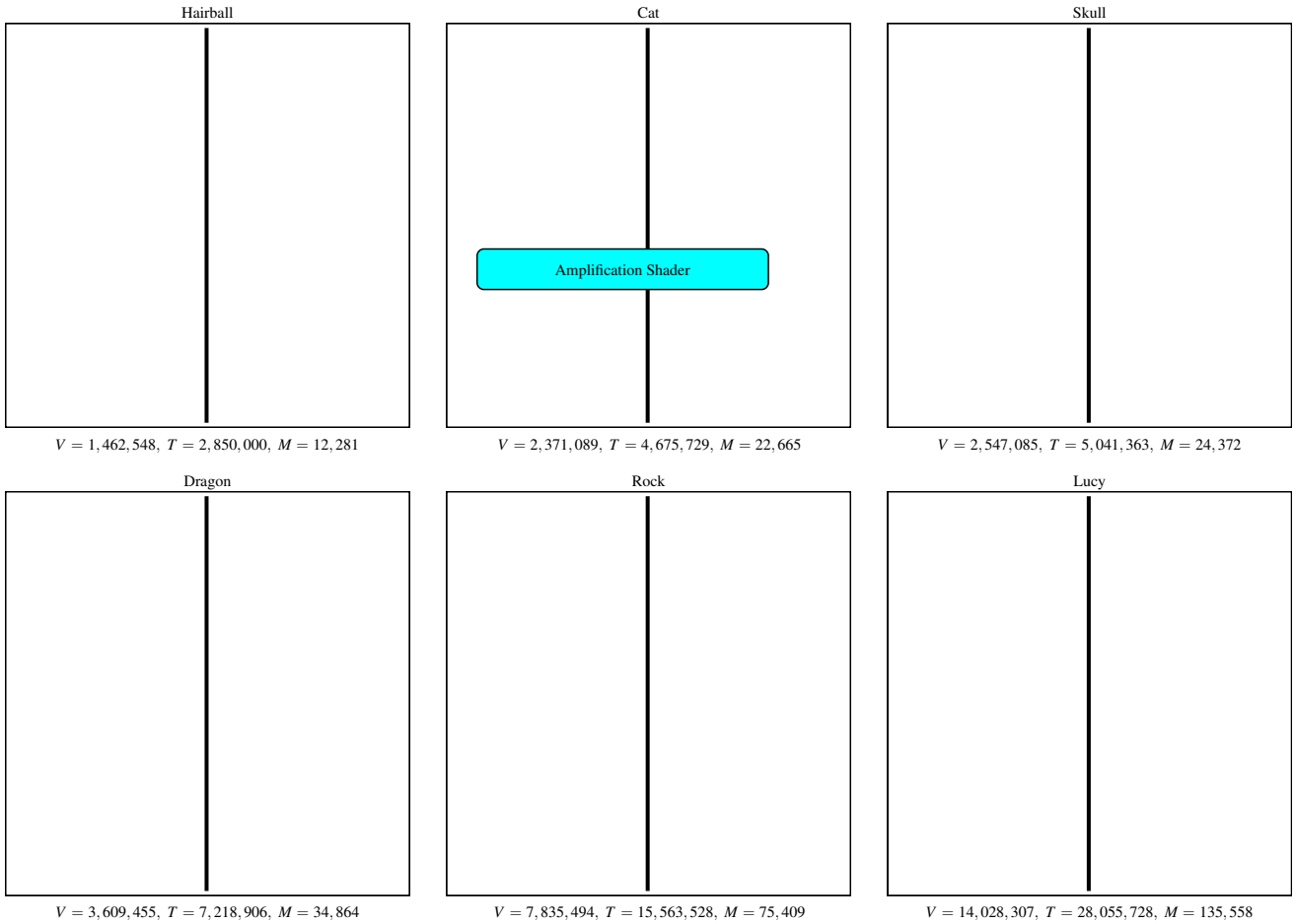


Figure 2: V denotes the number of vertices of the input mesh, T the number of triangles and M the number of meshlets coming from Meshoptimizer. Meshlets are visualized with randomized colors.

volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac

© 0x Eurographics - The European Association
for Computer Graphics and John Wiley & Sons Ltd.

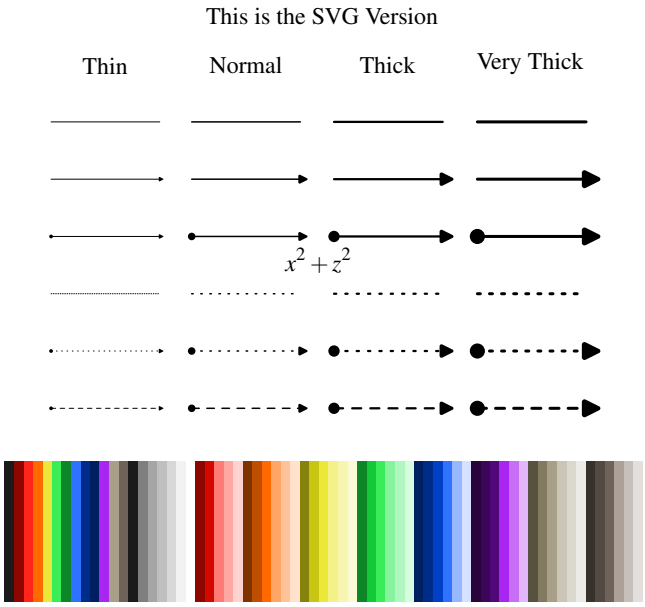


Figure 3: You can use Latex to render the fonts.

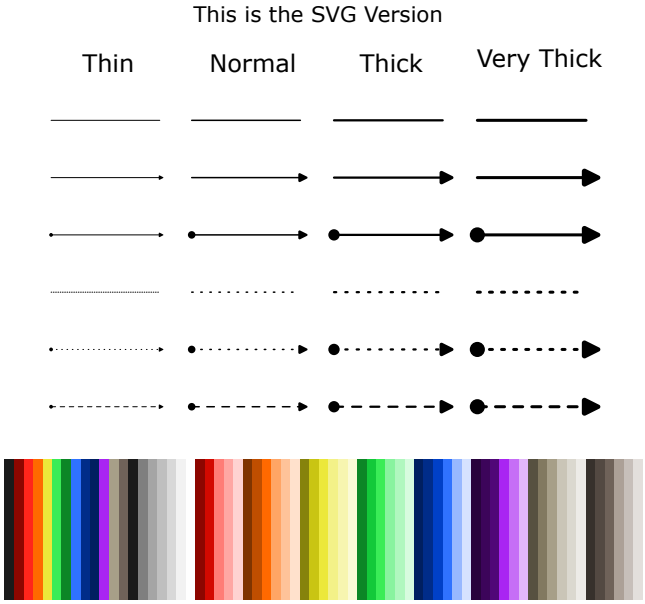


Figure 4: You can also use the fonts produced by the SVG file.

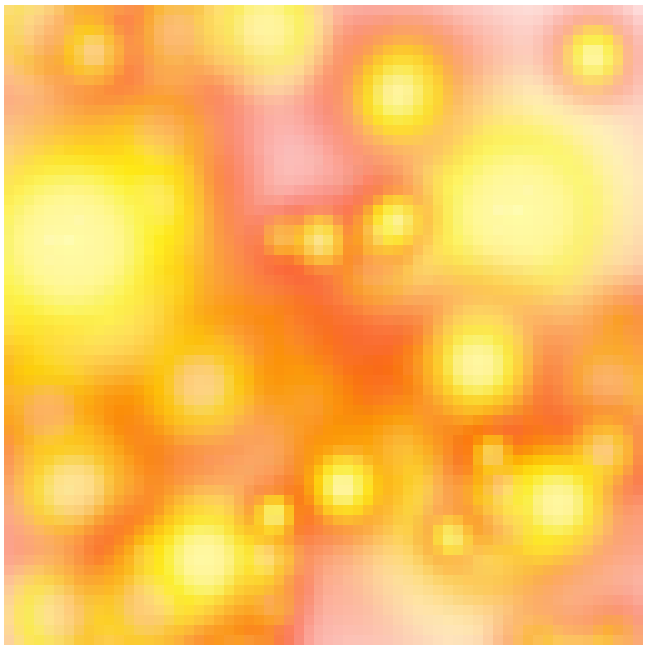


Figure 5: You can also directly include pixel graphics.

odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo. Aliquam vestibulum fringilla lorem. Sed neque lectus, consectetuer at, consectetuer sed, eleifend ac, lectus. Nulla facilisi. Pellentesque eget lectus. Proin eu metus. Sed porttitor. In hac habitasse platea dictumst. Suspendisse eu lectus. Ut mi mi, lacinia sit amet, placerat et, mollis vitae, dui. Sed ante tellus, tristique ut, iaculis eu, malesuada ac, dui. Mauris nibh leo, facilisis non, adipiscing quis, ultrices a, dui.

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetuer. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus

© 0x Eurographics - The European Association
for Computer Graphics and John Wiley & Sons Ltd.

scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi.

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetuer eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetuer tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim. Nunc purus neque, placerat id, imperdiet sed, pellentesque nec, nisl. Vestibulum imperdiet neque non sem accumsan laoreet. In hac habitasse platea dictumst. Etiam condimentum facilisis libero. Suspendisse in elit quis nisl aliquam dapibus. Pellentesque auctor sapien. Sed egestas sapien nec lectus. Pellentesque vel dui vel neque bibendum viverra. Aliquam porttitor nisl nec pede. Proin mattis libero vel turpis. Donec rutrum mauris et libero. Proin euismod porta felis. Nam lobortis, metus quis elementum commodo, nunc lectus elementum mauris, eget vulputate ligula tellus eu neque. Vivamus eu dolor.

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aenean nonummy magna non leo. Sed felis erat, ullamcorper in, dictum non, ultricies ut, lectus. Proin vel arcu a odio lobortis euismod. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Proin ut est. Aliquam odio. Pellentesque massa turpis, cursus eu, euismod nec, tempor congue, nulla. Duis viverra gravida mauris. Cras tincidunt. Curabitur eros ligula, varius ut, pulvinar in, cursus faucibus, augue.

Nulla mattis luctus nulla. Duis commodo velit at leo. Aliquam vulputate magna et leo. Nam vestibulum ullamcorper leo. Vestibulum condimentum rutrum mauris. Donec id mauris. Morbi molestie justo et pede. Vivamus eget turpis sed nisl cursus tempor. Curabitur mollis sapien condimentum nunc. In wisi nisl, malesuada at, dignissim sit amet, lobortis in, odio. Aenean consequat arcu a ante. Pellentesque porta elit sit amet orci. Etiam at turpis nec elit ultricies imperdiet. Nulla facilisi. In hac habitasse platea dictumst. Suspendisse viverra aliquam risus. Nullam pede justo, molestie nonummy, scelerisque eu, facilisis vel, arcu.

Appendix A: Implementation

Listing 1: Here is a longer description

```
import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int)  #dummy
        variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

                if M is None:
                    M = np.copy(VT)
                else:
                    M = np.concatenate((M,
                        VT), 1)

                VT = np.zeros((n*m,1), int)

    return M
```

Acknowledgments

We thank Dominik Baumeister. Meshes are courtesy of Aixterior (Rock, Skull), Morgan McGuire (Hairball), the Stanford 3D Scanning Repository (Dragon, Lucy), and Ekaterina Kozemerchak (Cat).

© 0x Eurographics - The European Association
for Computer Graphics and John Wiley & Sons Ltd.

References

[AG03] ALLIEZ P., GOTSMAN C.: Recent advances in compression of 3D meshes. In *Advances in Multiresolution for Geometric Modelling* (2003), Springer Berlin Heidelberg. 1

[MLDH15] MAGLO A., LAVOUÉ G., DUPONT F., HUDELOT C.: 3d mesh compression: Survey, comparisons, and emerging trends. *ACM Comput. Surv. 47*, 3 (2015). 1

[MSS24] MLAKAR D., STEINBERGER M., SCHMALSTIEG D.: End-to-end compressed meshlet rendering. *Computer Graphics Forum 43*, 1 (2024). 1

[PKJ05] PENG J., KIM C.-S., JAY KUO C.-C.: Technologies for 3D mesh compression: A survey. *Journal of Visual Communication and Image Representation 16*, 6 (2005). 1

© 0x Eurographics - The European Association
for Computer Graphics and John Wiley & Sons Ltd.